INCH-POUND

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MILITARY SPECIFICATION

COVERALLS, RADIOACTIVE CONTAMINANTS, PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of one-piece radioactive contaminant protective coverall made of polyester and cotton twill cloth.

1.2 Classification. The coveralls shall be in the following sizes (see 6.2):

Small - Medium - (Up to 43)
Large - X-Large - (44 to 48)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
JJ-S-746	- Stockinet, Surgical
JJ-W-155	- Webbing, Textile (Cotton, Elastic)
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
DDD-L-20	- Label: for Clothing, Equipage and Tentage (General Use)

MILITARY

MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-G-16491	- Grommet, Metallic
MIL-B-17757	- Boxes, Shipping, Fierboard (Modual Sizes)
MIL-F-21840	- Fastener Tapes, Hook and Loop, Synthetic
MIL-C-21851	- Cloth, Twill, Polyester and Cotton

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
* MIL-STD-2073-1	- Department of Defense Materials Procedures for Development and Application of Packaging Requirements
* MIL-STD-2073-2	- Packaging Requirement Codes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. POSTAL SERVICE MANUAL

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001).

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* 2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material. (see 6.4)

* 3.3.1 Basic material. The basic material shall be a polyester and cotton twill cloth, shade yellow 3511 conforming to MIL-C-21851, standard sample piece number 84058.

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3.3.1 Thread, cotton. The cotton thread used for seaming and stitching shall be ticket nos. 30 and 50, 3 ply, Type 1A3 of V-T-276. The color of the thread shall approximate the base fabic.

3.3.3 Webbing, elastic. The webbing shall be cotton, elastic, unbleached or bleached white conforming to the types and classes of JJ-W-155 as follows or a woven or braided elastic webbing of equivalent grade may be used:

Type	Class	Width	Use
I	10	1-1/2 inch	Back waist take-up
I	7	1 inch	Foot stirrups
II	1	5/16 inch	Hem at leg bottom

3.3.4 Wristlets, knitted, cotton. The knitted cotton wristlets shall be bleached or unbleached, 3 inches wide, conforming to type I of JJ-S-746.

3.3.5 Braid, nylon. The drawcord for the top of the coveralls shall be braided of multifilament nylon, and shall conform to the requirements specified in Table I when tested as specified in 4.4.1. The finished braid shall be not less than 3/16 inch in width and the color shall be white.

Table I - Physical characteristics

Characteristic	Requirements
Yarn denier	840 denier
Number of carriers	16
Ends per carrier	3
Picks per inch	11 (min.)

3.3.6 Grommets and washers. The grommets and washers shall conform to type I, class 2, size 1-J of MIL-G-16491.

3.3.7 Fastener tapes. The hook and loop fastener tapes shall conform to type II, class 1 of MIL-F-21840.. White 3054 tape fasteners shall be used in "small-medium" size coveralls and Red 3852 tape fasteners in "large-X-large" size coveralls to provide ready size identification. Tapes shall be as follows:

Width	Tape	Use
5/8 inch	Hook	Pocket flap
1 inch	Loop	Pocket
2 inch	Loop	Center back

3.3.8 Labels.

* 3.3.8.1 Combination label. Each coverall shall have a combination identification and instruction label comforming to type VI, class 15 of DDD-L-20. The printing shall show colorfastness to laundering. The inscription shall be as follows:

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COVERALLS, RADIOACTIVE CONTAMINANTS, PROTECTIVE
FIBER CONTENT: 65% POLYESTER AND 35% COTTON
CONTRACT NO.: DLA-100-00-0-0000 (Example)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (if other than contractor)

LAUNDER IN ACCORDANCE WITH PRESCRIBED REQUIREMENTS

* 3.3.8.2 Size label. The size label shall conform to type VI, class 2 of DDD-L-20 and shall read as follows:

SIZE: Small-Medium, up to size 43 (Example)
STOCK NO.: 8415-00-000-0000 (Example)

3.3.9 Radiation Symbol. The marking for the radiation symbol shall be in accordance with type IV of DDD-L-20. The radiation symbol shall be 3-3/4 inches in diameter conforming to the design and dimensions as shown in Figure 2.

3.4 Design. The coveralls, as shown in Figure 1, shall be one piece having full length sleeves with knitted wristlets, a two-section left breast pocket with abutted flaps, and a channeled nylon drawcord at neck closure. It shall incorporate elastic webbing for back waist take-up, for drawing in around leg bottoms, and foot stirrup straps. A black radiation symbol shall be stamped on upper left back of coverall.

3.4.1 Figures. The figures are furnished for information purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

* 3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are permitted on the working patterns. The patterns provide a seam allowance of 1/2 inch for all regular double lapped, double stitched seams.

3.5.1. Pattern parts. The component parts of the coveralls shall be cut from material as specified and in accordance with the following pattern parts:

Material	Nomenclature of pattern parts	Cut parts
Cloth, polyester/ cotton twill	Front - Back	2
	Front	2 1/
	Back	2 1/
	Sleeve, front	2
	Sleeve, back	2
	Pocket, double (dosimeter and film badge)	1

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Pocket flap (dosimeter)	1
Pocket flap (film badge)	1

1/ If the width of the material does not allow for economical cutting, the size Large - X-Large patterns, which provide for a four-piece body, may be used.

* 3.6 Stitches, seams and stitchings. The stitches, seams, and stitching types specified in Table II shall conform the FED-STD-751. Whenever two or more methods for seam or stitch type are given for the same operation, any one may be used. Where stitch type 401 is used, the looper (under thread) shall be on the inside of the coveralls. Ends of all seams produced by stitch type 401, shall be caught in other seams or stitchings. Unless otherwise specified, all double needle seaming shall be 1/4 inch gauge with the row of stitching next to the folded edge not more than 1/16 inch from edge. The seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats or open seams will result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Edgestitching and top stitching shall be 1/16 inch from folded edge. Hook and loop fastener tape stitching shall be 1/16 to 1/8 inch from the edge of the tape. When 2 inch wide fastener tape is used, an extra row of stitching shall be down the middle of the length. The bight of the overedge stitching shall be 3/16 inch. The guide and knife shall be set to trim only the ravelled ends of the fabric.

3.6.1 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table II.

* 3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching (except label) shall be overlapped not less than 1/2 inch. The ends of the label stitching shall be overlapped not less than 3 stitches. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stithces or 401 stitch type thread breaks may be repaired by using 301 stitch type. When stitch type 401 is permitted as an option, it shall not be used unless the ends are caught in other seams or stitching.

* 3.7 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with the operation requirements specified in Table II. The contractor is not required to follow the exact sequence of operations listed provided that the finished coveralls are identical to those produced by following the sequence of operations as listed in Table II.

3.7.1 Shade and size marking. The component parts of the coveralls shall be marked, ticketed, or bundled to insure a uniform shade and size throughout the garment. Any method may be used except:

1. Corrosive metal devices.

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2. Sew-on shade tickets.
3. Adhesive type tickets which discolor or adhere to the material upon removal of tickets.

NOTE: The use of ink pad numbering machine, rubber stamp or pencil is allowed, provided the numbering does not show on the outside of the garment and wherever possible, is covered by the seam allowance.

* 3.7.2 Abbreviations in table of operations. The abbreviations used in Table II are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
smlr	-	similar
Btnhl	-	Buttonhole
incl	-	including
dbl	-	double
chnstch	-	chainstitch

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TABLE II - CONSTRUCTION OF COVERALL

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
1.	Cutting a. The coverall shall be cut in strict accordance with patterns furnished, which shall show size, placement of component parts, directional lines for cutting, and notches for proper assembly of all parts. Directional lines shall be placed in the warp direction of the material and may vary from the warp direction by not more than 1-1/2 inches. Measurements shall be taken at the top and bottom of the directional line on patterns to the selvage edge of the fabric on the even side of the lay. The difference between the two measurements shall not exceed 1-1/2 inches.					
	b. All component parts shall be cut from one piece of material. The filling flush side shall be the face and shall finish as the exposed surface of the coveralls.					
2.	Replacement of defective components During the spreading, cutting and manufacturing process, component parts having material defects or damages that are classified as defects in section 4 shall be removed from production and replaced with non-defective and properly matched components.					
3.	Marking All parts shall be marked, bundled or ticketed to insure proper assembly throughout the coverall (see 3.7.1).					
4.	Pocket (dosimeter and film badge) a. Hem top of pocket with edge turned under 1/4 inch. Hem shall measure 5/8 (± 1/8) inch wide.	301	EFb-1	10-12	30	30
	b. Position 1 inch wide loop fastener tape 1/4 inch from folded top edge of pocket across full width of finished pocket. Stitch on all four sides 1/16 to 1/8 inch from edge.	301	LSbj-1	10-12	30	30
	c. Double stitch 1/4 inch gauge turning under the sides and bottom edge of pocket 3/8 inch, placing on left front, according to pattern marks.	301	LSd-2	10-12	30	30

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TABLE II - CONSTRUCTION OF COVERALL

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	d. Double stitch 1/4 inch gauge on pocket, according to marks on pattern, to form separate dosimeter and film badge sections. All stitching shall cross loop fastener tape.	301	SSv-2	10-12	30	30
	e. The inside measurements for the finished pocket sections shall be as follows: Dosimeter 2 (± 1/8) inches wide and 4-3/4 (± 1/4) inches deep. Film badge 2-5/8 (± 1/8) inches wide and 3-1/2 (± 1/4) inches deep.					
5.	Pocket flaps					
	a. Fold flap, face to face, and stitch sides 1/4 inch from edge. Turn to finished position and work out points.	301	SSa-1	10-12	30	30
	b. Position 5/8 inch hook fastener tape to underside of pocket flaps full width of finished flaps, and stitch 1/16 to 1/8 inch from edge on all four sides.	301	LSbj-1	10-12	30	30
	c. Stitch underside of pocket flap to left front according to marks on pattern, 1/4 inch from raw edge. Turn to first position and stitch 3/16 to 1/4 inch from folded edge. There shall be no space between flaps.	301	LSbl-2	10-12	30	30
	d. The finished dosimeter pocket flap shall measure 2-5/8 (± 1/8) inches wide and 2-1/8 (± 1/8) inches deep.					
	e. The finished film badge pocket flap shall measure 3-1/8 (± 1/8) inches wide and 2-1/8 (± 1/8) inches deep.					
6.	Front-Back					
	a. Double stitch center seams of fronts and backs with lefts lapped over rights.	301 or 401	LSc-2	10-12	30 30	30 50
	b. For 4-piece construction: Double stitch right and left fronts to right and left backs respectively with fronts lapped over backs.	301 or 401	LSc-2	10-12	30 30	30 50

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TABLE II - CONSTRUCTION OF COVERALL

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	c. Cut elastic webbing 15 inches long. Position webbing on backs to correspond to marks on pattern and stitch with 4 equidistant rows of stitching along webbing. Stitch ends of webbing 1/16 to 1/8 inch from ends.	301	SSau-4	10-12	30	30
7.	Sleeves					
	a. Double stitch the top seam and bottom seam of sleeve with the fronts lapped over the backs.	301 or 401	LSc-2	10-12	30 30	30 50
	b. Cut wristlet 8-1/2 inches long and fold in half lengthwise. Overedge stitch the end of sleeve to knit wristlet. The finished wristlet shall measure 4 (± 1/4) inches long.	503 504 or 602	SSa-1	8-10	50	50
8.	Armholes					
	Double stitch the sleeves to armholes with body lapped over sleeves.	301 or 401	LSc-2	10-12	30 30	30 50
9.	Inseams					
	Double stitch the leg inseams in one continuous operation with the fronts lapped over the backs.	301 or 401	LSc-2	10-12	30 30	30 50
10.	Drawcord channel - top hem					
	a. Secure grommet and washer approximately 1-1/2 (± 1/16) inches from top raw edge, centered on the center front seam.					
*	b. Turn under 1/4 inch top edge, fold over to make 3/4 (± 1/8) inch wide hem and stitch 1/16 inch from bottom folded edge, inserting size label approximately centered over the back joining seam.	301	EFb-1	10-12	30	30
	c. Insert drawcord into hem, allowing for 10 inches of free end of cord and knot the ends so as not to pull out of grommet.					
11.	Hanger loop					
	a. The loop shall be made by folding a piece of the basic fabric to the center with the edges abutting and stitched with a covering stitch on the underside.	406	EFh-1	10-12	50	50
	OR					

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TABLE II - CONSTRUCTION OF COVERALL

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	b. Fold hanger stripping in half with the raw edges turned in and stitch 1/16 to 1/8 inch from each edge.	301	EFp-2	10-12	30	30
	c. Stitch each end of the hanger loop to the inside back of coverall centered over the center back seam with 3/8 inch vertical bartacks, simultaneously catching the channeled nylon drawcord.	Brtck		28 per brtck	50	50
	d. Hanger loop shall measure finished 1/4 to 3/8 inch wide and shall have a free opening of 2-1/2 to 2-3/4 inches.					
12.	Elastic loop - leg hem					
	a. Cut elastic webbing 11-1/2 inches long. Backstitch or bartack across the width to form continuous loops 11 inches in circumference with one end of the elastic webbing overlapped 1/2 inch.	301 or Brtck	LSa-1	10-12 28 per brtck	30 50	30 50
	b. Make a 3/4 (± 1/8) inch hem on each leg bottom with the raw edge turned under 1/4 to 3/8 inch. Stitch 1/16 inch from inside folded edge simultaneously inserting the formed elastic loop without catching it in the stitching.	301	EFb-1	10-12	30	30
13.	Foot stirrup straps Cut elastic webbing 10-1/2 (± 1/4) inches long. Position ends adjacent to inside leg hem inseam and on opposite side of leg. Stitch, 1/4 inch from raw edge, two 3/8 inch long bartacks at each end of strap along hem seam line.	Brtck		28 per brtck	50	50
14.	Hood anchor point Position 2 by 4 inch loop fastener tape to center back of coveralls as indicated by marks on patterns. Stitch 1/16 to 1/8 inch from edge all four sides of tape.	301	LSbj-1	10-12	30	30

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TABLE II - CONSTRUCTION OF COVERALL

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
15.	Label Position the top edge of the combination identification and instruction label on the inside of the coverall adjacent to the folded edge of drawcord channel hem and centered over the back joining seam. Stitch the label on all four sides 1/16 inch from edge. Top edge of the label may be caught when performing operation 10b.	301	LSbj-1	10-12	30	30
16.	Radiation symbol Stamp radiation symbol, as indicated in Figure 2 on the upper left back section of the coverall midway between the center back and sleeve-armhole seams with the bottom of the symbol aligned with the base of the armhole.					
17.	Cleaning a. Trim all thread ends. NOTE: Care shall be taken when trimming thread ends that damage to the fabric does not result. b. Remove all loose threads. c. Remove all spots and stains.					

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3.8 Sizes and measurements. Unless otherwise specified, finished measurements of coveralls shall be as listed in Table III. All measurements and tolerances are expressed in inches.

Table III - Sizes and measurements

	Small-Medium	Large-X-Large	Tolerance
(A) Chest	53	58	± 1
(B) Seat	54	58	± 1
(C) Leg inseam	31	33	± 1/2
(D) Sleeve inseam	25	25	± 1/2
(E) Back length	64-1/2	70	± 1/2

NOTE: Refer to Figure 1 for A, B, C, D, and E measurements. The above measurements shall be taken with the coverall laid out smooth and flat and the neck opening fully extended.

(A) Chest - Twice the measurement taken across front at base of armholes from folded edge to folded edge.

(B) Seat - Twice the measurement taken across front six inches above the crotch seam from folded edge to folded edge.

(C) Leg inseam - Measurement shall be taken from junction of seams (crotch) along inseam to bottom of hemmed coverall legs.

(D) Sleeve inseam - Measurement shall be taken from the base of the armhole along inseam to bottom folded edge of knitted cuff.

(E) Back length - Total back length from bottom of leg hem to top of hem at neck opening - four inches from center seam.

3.9 Workmanship. The finished coveralls shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2 shall be inspected for compliance with design, construction, workmanship, appearance, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. The contractor shall furnish a certificate of compliance for the physical requirement of the nylon braid specified in 3.3.5.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.3. The sample shall be one finished pair of coveralls. The lot size shall be expressed in units of coveralls.

4.4.2.1 Visual examination. The coveralls shall be examined for defects in color, design, material, construction, workmanship, and marking and the defects classified in accordance with the list below:

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Defect	Classification Major	Minor
I. MATERIAL DEFECTS AND WORKMANSHIP DAMAGES		
a. Any weakening material defect, such as a smash, hole, cut, tear, burn, loose slub, mend, thin place, needle chew or drill hole	X	
b. Any outside part shaded		X
c. One or more broken or missing yarns	X	
d. Any nonweakening material defect such as knot, misweave, unsightly slub, loose yarn, coarse yarn, area of no dye penetration, dye streak or woven in waste		X
II. CLEANNESS		
a. Any spot or stain on outside		X
b. Thread ends not trimmed to 1/4 inch or loose threads not removed (to be scored only when condition exists on major portion of garment)		X
c. Two or more shade or size tickets not removed		X
d. Shade marking exposed and visible on outside of coverall		X
III. SHADED PARTS		
* a. Shade of coverall other than specified	X	
* b. Any outside part shaded		X
* c. Shade of thread markedly in contrast with shade of coverall		X
IV. CUTTING		
Any part not cut in accordance with directional lines indicated on patterns or in accordance with specification requirements	X	
V. COMPONENT AND ASSEMBLY		
a. Any component part omitted or required operation omitted (unless otherwise classified herein)	X	
b. Any component part twisted, distorted, pleated, tight, or full		X
c. Any component not as specified (type, size, color, etc)	X	
d. Any operation not as specified (unless otherwise classified herein)		X
VI. SEAMS AND STITCHING		
a. Accuracy of seaming		
1. Seam puckered, irregular, wavy or pleated (unless otherwise classified herein)		X

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Defect	Classification Major	Minor
2. Part of coverall caught in any unrelated stitching operation	X	
3. Permanent fold or pleat in cloth (caught in seaming or stitching)		X
4. End of stitching when not caught in another seam or stitching backstitched less than 3/8 inch		X
5. Ends of a continuous line of stitching overlapped less than 1/2 inch		X
6. Color of thread used on outside not approximating color of base fabric		X
7. Needle perforations visible on outside		X
* 8. Thread breaks not secured or stitched back of the break less than 1/2 inch		X
b. Gage of stitching and seam allowance		
1. Gage of stitching irregular, not uniform or not within range specified by 1/16 inch or more (score when condition occurs on more than half the length of seam)		X
2. Edge of top stitching sewn too close to edge, resulting in damage to cloth		X
3. Width of seam allowance not as specified by 1/16 inch or more (score when condition occurs on more than half the length of seam)		X
c. Open seam 1/		
1. 3/4 inch or less		X
2. More than 3/4 inch	X	
d. Runoff, skipped or broken stitches		
1. On joining seam score as open seam 1/		
2. Edge or top stitching (not resulting in an open seam)		X
e. Raw edges		
1. On outside:		
a. 3/4 inch or less		X
b. more than 3/4 inch	X	
2. On inside:		
More than 1 inch		X

1/ A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped or runoff stitches occur. On double stitched seams a seam is considered open when one or both stitches of the seam is opened.

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Defect	Classification Major	Minor
f. Seam and stitch type		
1. Not specified seam or stitch type		X
2. Looper thread of 401 stitch type finishing on the outside of the coverall		X
g. Stitch tension		
1. Loose tension resulting in a loose seam	X	
2. Loose tension (exposed loops of top or bottom threads)		X
3. Tight tension (stitches break when normal strain is applied in direction of the seam or stitching)	X	
h. Stitches per inch (score only when condition exists on more than half the length of the seam)		
1. Less than minimum specified		X
2. Any number in excess of the maximum specified		X
i. Direction of seam lap		
Any seam not lapped in the direction specified		X
j. Crotch seam		
Crotch seam and seat seam staggered more than 1/2 inch (measured from center to center of seams)		X
VII. ELASTIC LEG LOOP AND HEM		
a. Loop not continuous	X	
b. Loop caught in hem stitching	X	
c. Loop less than 11 inches in circumference		X
d. Hem less than 5/8 or more than 7/8 inch in width		X
VIII. ELASTIC WEBBING (BACK)		
a. Other than 4 rows of stitching	X	
b. Not stitched under tension to 24 inch length	X	
c. Other than specified width or length		X
IX. FOOT STIRRUP STRAPS		
a. Bartack missing, insecure, misplaced not serving intended purpose	X	
b. Misplaced by more than 1/4 inch	X	
c. Other than specified width or length		X

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Defect		Classification Major	Minor
X.	FASTENER TAPE		
a.	Misplaced by more than 1/4 inch	X	
b.	Foreign matter (thread, etc.) trapped in hook tape		X
c.	Other than type, length or width specified	X	
d.	Other than color required for specified size		X
XI.	GROMMET		
a.	Broken, or not securely clinched to fabric	X	
b.	Any sharp or rough edge	X	
c.	Too tightly clinched cutting fabric	X	
d.	Not on center front seam		X
e.	Does not provide clear access into hem	X	
XII.	HANGER LOOP		
a.	Off center with center of back seam by 3/4 inch		X
b.	Free opening under 2-1/2 inches or over 2-3/4 inches		X
c.	Width of hanger loop under 1/4 inch or over 3/8 inch		X
XIII.	DRAWCORD		
a.	Omitted	X	
b.	Not caught in loop bartacks	X	
XIV.	DRAWCORD CHANNEL		
a.	Channel hem less than 5/8 or more than 7/8 inch wide		X
b.	Cord ends not knotted to form a continuous loop		X
c.	Free end of drawcord less than 10 inches in length with neck opening fully extended		X
d.	Drawcord caught in hem stitching	X	
XV.	POCKETS		
a.	Not positioned properly by 1/2 inch or more		X
b.	Dosimeter pocket (inside measurement):		
1.	Less than 4-1/2 inches deep	X	
2.	More than 5 inches deep		X
3.	Less than 1-7/8 inches wide	X	
4.	More than 2-1/8 inches wide		X
c.	Film badge pocket (inside measurement)		
1.	Less than 3-1/4 inches deep	X	
2.	More than 3-3/4 inches deep		X
3.	Less than 2-1/2 inches wide	X	
4.	More than 2-3/4 inches wide		X

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Defect	Classification Major	Minor
XVI. POCKET FLAPS (dosimeter and film badge)		
a. Two flaps not abutted or out of alignment		X
b. One or both flaps less than 2 inches or more than 2-1/4 inches deep		X
c. Dosimeter flap less than 2-1/2 inches or more than 2-3/4 inches wide		X
d. Film badge flap less than 3 inches or more than 3-1/4 inches wide		X
XVII. SLEEVE		
Sleeve pleated or puckered at armhole		X
XVIII. KNITTED WRISTLET		
a. Not type specified		X
b. Less than 3-3/4 inches or more than 4-1/4 inches in length		X
c. Not double thickness	X	
XIX. LABELS		
* a. Omitted, incorrect or illegible	X	
* b. Misplaced		X
XX. RADIATION SYMBOL		
* a. Omitted, incorrect or illegible	X	
* b. Not placed according to specification	X	

4.4.2.2 Size measurement defects.

a. Any measurement deviating from dimensions and tolerances specified in Table III shall be scored as a size measurement defect.

b. Sleeve and leg inseam lengths uneven by 1/2 inch or more shall be scored as a size measurement defect.

4.4.3 Acceptable quality levels and inspection levels. The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

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	Inspection Level	AQL
For visual examination in 4.4.2.1:		
Major	II	2.5
Total (Major and Minor combined)	II	15.0
For measurement examination in 4.4.2.2:		
One class defect	S-3	4.0

4.4.4 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5 defects per hundred units in accordance with MIL-STD-105.

Examine	Defect
Marking (exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

4.4.5 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

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Examine	Defect
Finished dimensions	Length, width,or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation packaging. Preservation packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each coverall shall be folded as follows:

(1) Lay coverall flat, face down.
(2) Fold sleeves over coverall so that outseams meet at the center seam or back.
(3) The outer edges of legs and body shall be adjusted by folding so that width of coverall measures approximately 14-1/2 inches.
(4) Fold coverall in half by bringing legs up to the top.
(5) Fold coverall in thirds by bringing the bottom up and the top down with the combination label visible for ready identification. The completely folded coverall shall measure approximately 11-1/2 by 14-1/2 inches.
(6) Each folded pair of coveralls shall be individually packaged in a clear, snug-fitting polyethylene film bag of 0.0015 (± 20% tolerance) inch thickness. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. As an alternative, the polyethylene bag may be of the tuck-in or reverse flap type, where heat seal closure and corner hole are not required.

5.1.2 Level C. Coveralls shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Twenty coveralls, of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container, assembled, closed, and reinforced conforming to type CF, class weather-resistant, grade V15c, variety DW, size 3A of MIL-B-17757. The liner shall be type CF, class domestic, conforming to MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and ten in depth within the shipping container. The contents of each container shall be completely covered on top and bottom with a sheet of commercial grade kraft paper.

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* 5.2.2 Level B. Twenty coveralls, of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF, class domestic, grade 275, variety DW, size 3A of MIL-B-17757. The liner shall be type CF, class domestic in accordance with the requirements specified in MIL-B-17757. Level A packages shall be packed flat, two in length, one in width and ten in depth within the shipping container. The contents of each container shall be completely covered on top and bottom with a sheet of commercial grade kraft paper.

5.2.3 Level C. Coveralls packaged, as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers, palletized unit loads shall be marked in accordance with MIL-STD-129.

* 5.4 Palletization. When specified (see 6.2) item packed as specified shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K, and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, Type IV; Type V, Class 1, Size 2; or Type VIII, fabricated from wood group I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from wood groups specified in MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

* 5.5 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The coveralls are intended to be used as a protective garment by personnel working in low level radioactive contaminated areas in conjunction with the hood, gloves, and footwear items of the related radioactive contaminants protective outfit.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Sizes required (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, testing and approval of the first article.
e. Levels of packaging and packing required (see 5.1 and 5.2)
f. When palletization is required (see 5.4)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.5).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids (see 3.1 and 3.5).

6.4 Recycled materials. It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Subject term (key word) listing.

Protection garment, low level radiation
Protection garment, polyester/cotton twill
Protection garment, unit of RCP outfit

Custodian:
Navy - NU
Air Force - 99

Preparing Activity:
Navy - NU

Review Activity:
DLA - CT
Navy - SH, AS
Air Force - 82

Project No. 8415-0682

User Activities:
Navy - OS, CG
Air Force - 45

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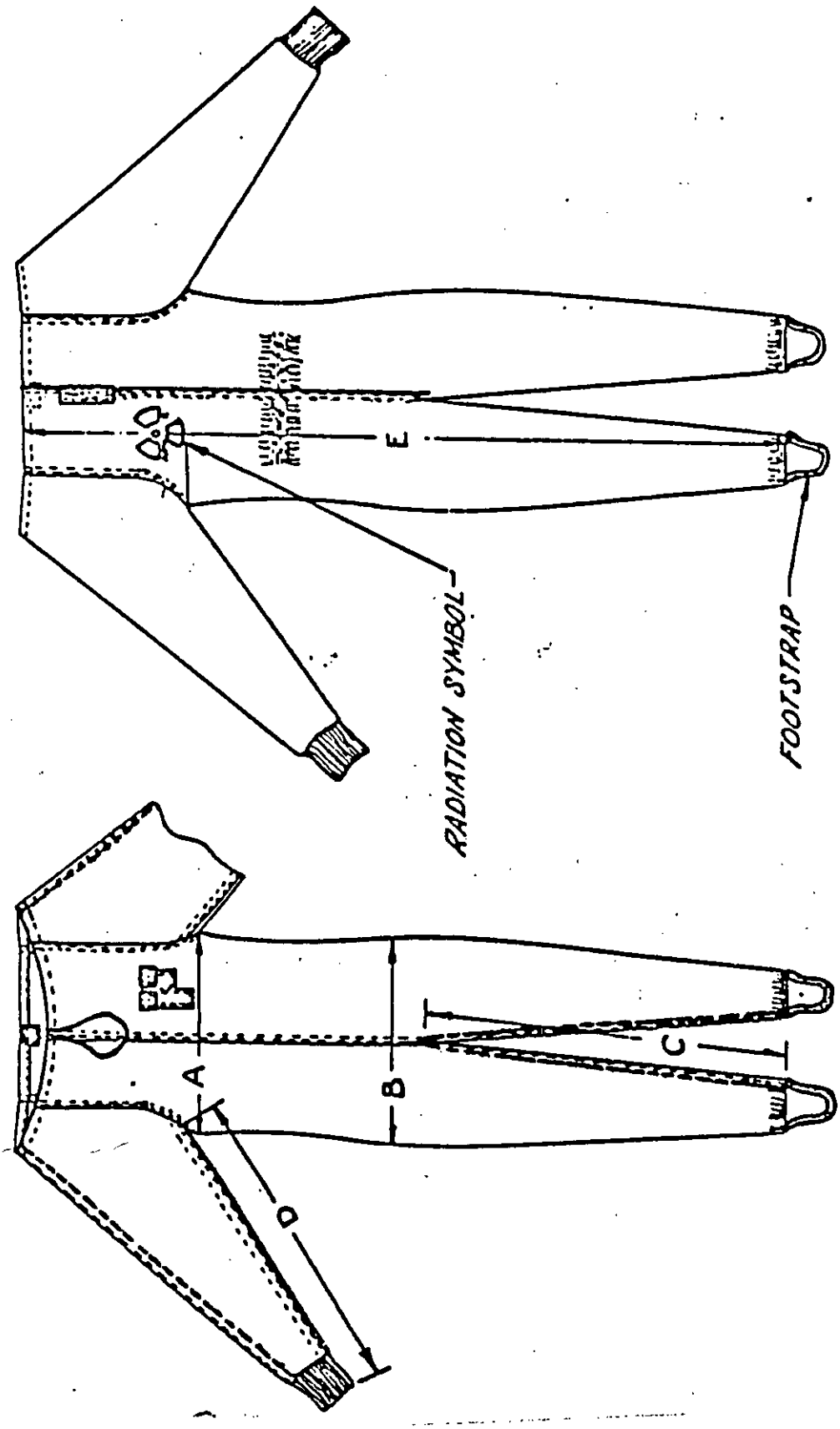


FIGURE 1 COVERALLS, RADIOACTIVE CONTAMINANTS, PROTECTIVE

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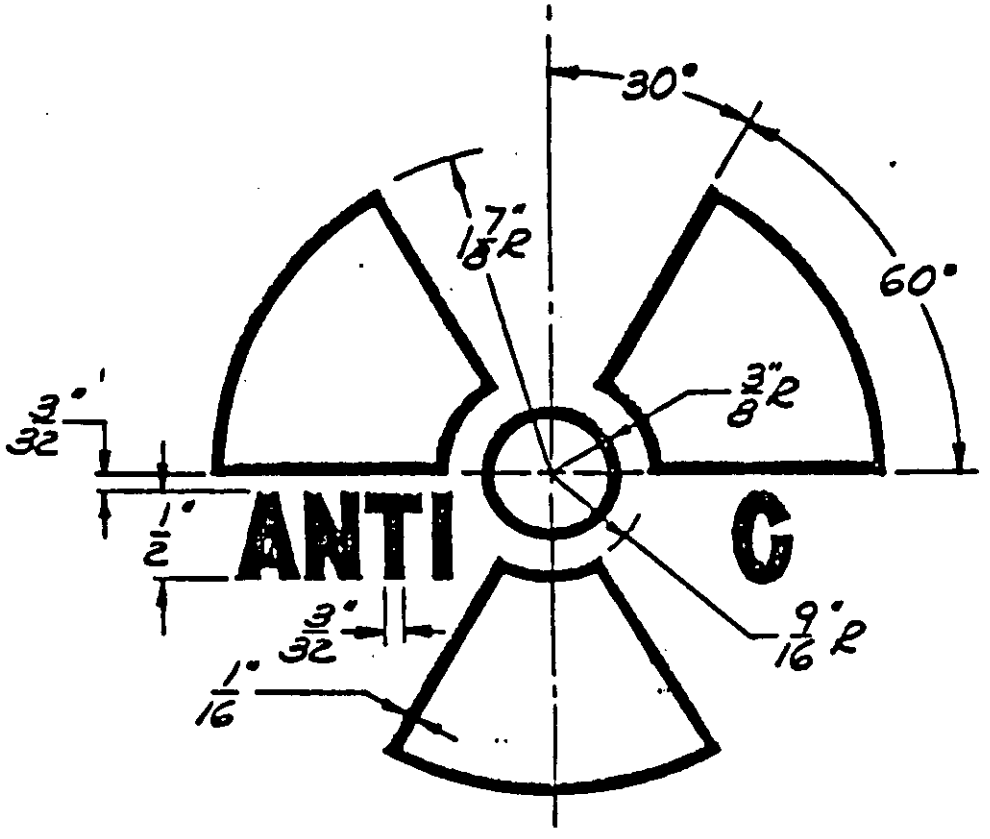


FIG. 2 RADIATION SYMBOL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-82243E	COVERALLS, RADIOACTIVE CONTAMINANTS, PROTECTIVE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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